

BIOL343 Coding Challenge 1

Group Members

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```
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```

Part 1 — Import and Inspect the Dataset

1. Import the Data

The dataset is located in the `data` folder and is named:

`A1_Data_Excel.csv`

Use `read.csv()` to import the file and save it as an object called `assn1_data`.

```
assn1_data <- read.csv("data/A1_Data_Excel.csv")
```

2. Inspect the Data

Use the following functions to examine the dataset:

- `class()`
- `dim()`
- `str()`
- `head()`
- `tail()`

```
# class()  
class(assn1_data)
```

```
## [1] "data.frame"
```

```
# dim()  
dim(assn1_data)
```

```
## [1] 48 8
```

```
# str()  
str(assn1_data)
```

```
## 'data.frame': 48 obs. of 8 variables:
## $ population_type : chr "A" "A" "A" "A" ...
## $ population_name : chr "MI M2" "MI M3" "MI M5" "MI M7" ...
## $ latitude : num 43.3 45.6 42.4 45.6 43.4 ...
## $ longitude : num -84.9 -84.7 -84 -84.7 -85.9 ...
## $ ovary_width : num 3.62 3.61 2.98 3.48 3.43 3.87 3.36 3.62 3.55 3.39 ...
## $ ovary_height : num 3.48 3.59 2.68 4.08 2.94 3.11 3.28 2.86 2.87 3.17 ...
## $ average_august_temperature: chr "21" "18.3" "20.8" "18.2" ...
## $ flower_number : num NA NA NA NA NA ...
```

```
# head()
head(assn1_data)
```

```
## population_type population_name latitude longitude ovary_width ovary_height
## 1 A MI M2 43.264 -84.884 3.62 3.48
## 2 A MI M3 45.615 -84.689 3.61 3.59
## 3 A MI M5 42.418 -84.013 2.98 2.68
## 4 A MI M7 45.623 -84.705 3.48 4.08
## 5 A MI M8 43.350 -85.935 3.43 2.94
## 6 A MI M9 45.600 -84.709 3.87 3.11
## average_august_temperature flower_number
## 1 21 NA
## 2 18.3 NA
## 3 20.8 NA
## 4 18.2 NA
## 5 20.1 NA
## 6 18.2 NA
```

```
# tail()
tail(assn1_data)
```

```
## population_type population_name latitude longitude ovary_width ovary_height
## 43 S ON T16 44.902 -75.870 3.34 3.11
## 44 S ON T3 43.208 -80.650 3.35 2.83
## 45 S ON T7 42.049 -83.106 3.78 3.18
## 46 S ON T8 42.257 -81.859 3.30 2.88
## 47 NA NA NA NA
## 48 NA NA NA NA
##
## 43
## 44
## 45
## 46
## 47
## 48
## flower_number
## 43 45.3
## 44 NA
## 45 NA
## 46 NA
## 47 NA
## 48 NA
```

3. Clean the Dataset

Based on your inspection, retain the appropriate rows and columns.

Save the cleaned dataset as `assn1_data_mod`.

```
assn1_data_mod <- assn1_data[1:46, 1:8]
```

4. Calculate Ovary Area

Create a new variable called `ovary_area` using:

$$\text{ovary area} = \text{ovary width} \times \text{ovary height}$$

```
assn1_data_mod$ovary_area <- assn1_data_mod$ovary_width * assn1_data_mod$ovary_height
```

Use `head()` to confirm that the new column was created.

```
# Check the first few rows.  
head(assn1_data_mod)
```

```
##   population_type population_name latitude longitude ovary_width ovary_height  
## 1                A             MI M2   43.264   -84.884         3.62         3.48  
## 2                A             MI M3   45.615   -84.689         3.61         3.59  
## 3                A             MI M5   42.418   -84.013         2.98         2.68  
## 4                A             MI M7   45.623   -84.705         3.48         4.08  
## 5                A             MI M8   43.350   -85.935         3.43         2.94  
## 6                A             MI M9   45.600   -84.709         3.87         3.11  
##   average_august_temperature flower_number ovary_area  
## 1                        21             NA    12.5976  
## 2                       18.3             NA    12.9599  
## 3                       20.8             NA     7.9864  
## 4                       18.2             NA    14.1984  
## 5                       20.1             NA    10.0842  
## 6                       18.2             NA    12.0357
```

Part 2 — Simulate and Combine Data

You will now create a small simulated dataset with the same variables as the original dataset.

For continuous variables such as ovary width and ovary height, generate values from a **normal distribution** using `rnorm()`. Use a mean and standard deviation similar to those in the previously loaded `assn1_data_mod` dataset. You can calculate these values from the original data using `mean()` and `sd()`.

Before generating any random values, use `set.seed()` so that your simulated results can be reproduced.

Use the seed value 343.

```
set.seed(343)
```

Run `set.seed(343)` before generating your simulated values. Using the same seed will produce the same random values each time the code is run.

5. Create Vectors

Create vectors for **10 new observations**.

Population Type

Create a vector called `sim_population_type` containing "A" and "S" values using `sample()` with size as 10 and replacement as `TRUE`

```
sim_population_type <- sample(
  assn1_data_mod$population_type,
  size = 10,
  replace = TRUE
)
```

Ovary Width

Create a numeric vector called `sim_ovary_width` containing 10 simulated ovary-width values.

Use `rnorm()` and choose the mean and standard deviation based on the ovary-width values in `assn1_data_mod`.

```
#calculate mean and sd of ovary-width

# mean
mean(assn1_data_mod$ovary_width, na.rm = TRUE)

## [1] 3.424565

# sd
sd(assn1_data_mod$ovary_width, na.rm = TRUE)

## [1] 0.2924054

sim_ovary_width <- sim_ovary_width <- rnorm(
  n = 10,
  mean = 3.42,
  sd = 0.29)
```

Ovary Height

Create a numeric vector called `sim_ovary_height` containing 10 simulated ovary-height values.

Use `rnorm()` and choose the mean and standard deviation based on the ovary-height values in `assn1_data_mod`.

```
#calculate mean and sd of ovary-height

# mean
mean(assn1_data_mod$ovary_height, na.rm = TRUE)

## [1] 3.025

# sd
sd(assn1_data_mod$ovary_height, na.rm = TRUE)

## [1] 0.3584395

sim_ovary_height <- rnorm(
  n = 10,
  mean = 3.025,
  sd = 0.3584395)
```

Flower Number

Create a numeric vector called `sim_flower_number` containing 10 simulated flower-count values.

```
#calculate mean and sd of flower_number

# mean
mean(assn1_data_mod$flower_number, na.rm = TRUE)

## [1] 58.0156

# sd
sd(assn1_data_mod$flower_number, na.rm = TRUE)

## [1] 44.13149

sim_flower_number <- sim_ovary_height <- rnorm(
  n = 10,
  mean = 58.0156,
  sd = 44.13149)
```

6. Create a Simulated Data Frame

Combine your four vectors into a data frame called `sim_data`.

Make sure the column names match the corresponding columns in `assn1_data_mod`.

```
sim_data <- data.frame(
  population_type = sim_population_type,
  ovary_width = sim_ovary_width,
  ovary_height = sim_ovary_height,
  flower_number = sim_flower_number
)
```

Calculate `ovary_area` for the simulated observations.

```
sim_data$ovary_area <- sim_data$ovary_width * sim_data$ovary_height
```

Use `str()` and `head()` to inspect your simulated dataset.

```
# str()
str(sim_data)

## 'data.frame': 10 obs. of 5 variables:
## $ population_type: chr "A" "S" "A" "S" ...
## $ ovary_width : num 3.36 3.28 3.58 3.11 3.19 ...
## $ ovary_height : num 124.3 68.5 83.1 -18.4 131.8 ...
## $ flower_number : num 124.3 68.5 83.1 -18.4 131.8 ...
## $ ovary_area : num 417.1 224.7 297.8 -57.2 420.9 ...

# head()
head(sim_data)

## population_type ovary_width ovary_height flower_number ovary_area
## 1 A 3.356681 124.26183 124.26183 417.10735
## 2 S 3.279707 68.52720 68.52720 224.74912
## 3 A 3.584167 83.08373 83.08373 297.78593
## 4 S 3.105412 -18.42261 -18.42261 -57.20981
## 5 S 3.192680 131.82666 131.82666 420.88038
## 6 S 3.568953 21.78647 21.78647 77.75490
```

7. Combine the Datasets

Combine the original cleaned dataset and the simulated dataset using `rbind()`.

Save the combined dataset as `combined_data`.

```
combined_data <- rbind(
  assn1_data_mod[, names(sim_data)],
  sim_data
)
```

Check the dimensions of the combined dataset.

```
# Check the dimensions.
dim(combined_data)
```

```
## [1] 56  5
```

Part 3 — Basic Summary Statistics and Visualization

8. Flower Number Summary Statistics

Calculate the following for `flower_number` in `combined_data`:

- Minimum
- Maximum
- Mean
- Median
- Standard deviation

```
# Minimum
min(combined_data$flower_number, na.rm = TRUE)
```

```
## [1] -18.42261
```

```
# Maximum
max(combined_data$flower_number, na.rm = TRUE)
```

```
## [1] 181.76
```

```
# Mean
mean(combined_data$flower_number, na.rm = TRUE)
```

```
## [1] 61.90953
```

```
# Median
median(combined_data$flower_number, na.rm = TRUE)
```

```
## [1] 52.59
```

```
# Standard deviation
sd(combined_data$flower_number, na.rm = TRUE)
```

```
## [1] 44.4753
```

9. Use `summary()`

Use `summary()` on both `ovary_area` and `flower_number`.

```
# Ovary area
```

```
summary(combined_data$ovary_area)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -57.210   9.498  10.708  52.462  12.667  420.880
```

```
# Flower number
```

```
summary(combined_data$flower_number)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  -18.42   28.76   52.59   61.91   86.14  181.76     21
```

11. Compare Population Types

Create two subsets from `combined_data`:

- `asexual_data` for population type "A"
- `sexual_data` for population type "S"

```
asexual_data <- subset(
  combined_data,
  population_type == "A"
)
```

```
sexual_data <- subset(
  combined_data,
  population_type == "S"
)
```

Calculate the mean ovary area for each group.

```
# Mean ovary area for A
mean(asexual_data$ovary_area, na.rm = TRUE)
```

```
## [1] 53.6535
```

```
# Mean ovary area for S
```

```
mean(sexual_data$ovary_area, na.rm = TRUE)
```

```
## [1] 51.42877
```

Question 1: Which population type has the larger mean ovary area?

Answer:

11. Visualization of ovary area

Use `ggplot2` to create a figure of your choice comparing ovary area between population types using `theme_classic`

Your figure should include:

- use `facet_wrap` or `facet_grid` if needed
- appropriate axis labels

- a descriptive title
- legends if necessary

```
library(ggplot2)
```

```
##boxplot is usually one for visualizing but they could also much scatter plot with facets or geom_hist
```